

Solⁿ of the free Dirac Eq:

$$(i\not{D} + m)\psi = 0 \Leftrightarrow p^\mu \gamma_\mu + m = 0$$

$$\text{and } p^2 + m^2 = 0$$

$$(\partial^2 - m^2)\psi = 0$$

$$\boxed{\psi(x) = u(p) e^{i p x}} \Rightarrow (i\not{p} + m)u(p) = 0$$

$$\text{Solutions: } u^s(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ \sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix} \quad s=1, 2$$

$$\xi^1 \equiv \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\xi^2 \equiv \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$* \psi \text{ is in } \mathbb{M}^4 \quad \text{so } u^s(p) \in \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

$$\sigma^\mu = (1, \sigma^i) \quad \bar{\sigma}^\mu = (1, -\sigma^i)$$

In the rest frame:

$$\vec{p} = 0, \quad p^0 = m \Rightarrow \sqrt{-p \cdot \sigma} = \sqrt{-p^0 \sigma^0} = \sqrt{m} 1$$

$$\rightarrow u^1(p) \sim \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \quad u^2 \sim \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

$$* \psi(x) = u(p) e^{i p x}, \quad p^2 + m^2 = 0, \quad p^0 > 0$$

$$(-i\not{p} + m)u(p) = 0$$

$$u^s(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi^s \\ \sqrt{p \cdot \bar{\sigma}} \xi^s \end{pmatrix}, \quad u^s(p) \sim \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} \text{ or } \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$$

$$* \text{Normalize } \xi^{\dagger} \xi = 1 \Leftrightarrow \xi^{\dagger r} \xi^s = \delta^{rs}$$

$$\Rightarrow \bar{u}^s = (u^s)^\dagger i \gamma^0 = (u^s)^\dagger i \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$= (u^s)^\dagger \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$= (a^s, b^s) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$= (b^s, a^s)$$

$$\bar{u}^s u^r = (b^s, a^s) \begin{pmatrix} a^r \\ b^r \end{pmatrix} = b^s a^r + a^s b^r$$

$$= \sqrt{p \cdot \sigma} \sqrt{p \cdot \bar{\sigma}} 2 \delta^{rs}$$

Where

$$p^\mu \sigma_\mu p^\nu \bar{\sigma}_\nu$$

$$\text{However: } \{\sigma^\mu, \bar{\sigma}^\nu\} = -2g^{\mu\nu}$$

...
Let's go Bosonic theory.